INCH-POUND

MIL-R-82817(OS)
29 December 1989
SUPERSEDING
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25 October 1983

MILITARY SPECIFICATION

REINFORCEMENT, CARBON

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for carbon reinforcement, referred to herein as the reinforcement.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9330

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1106, Chicago, IL 60606-5945.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C 135 True Specific Gravity of Refractory Materials by Water Immersion

ASTM D 1682 Breaking Load and Elongation of Textile Fabrics

ASTM D 1777 Measuring Thickness of Textile Materials

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ASTM D 3176 Ultimate Analysis of Coal and Coke

ASTM D 3776 Mass per Unit Area (Weight) of Woven Fabric

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the reinforcement shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 **Material.** The reinforcement furnished in accordance with this specification shall be an 8-harness, satin weave, rayon-base, carbon fabric processed at a temperature range of 2000 to 2850°F with properties as defined in table I.

TABLE I. Properties.

Material	Limits	
	Minimum	Maximum
Carbon, percent	94.0	100.0
Ash content, percent	0.0	1.0
Moisture, percent	0.0	3.0
Weight, ounces per square yard	7.0	9.5
Breaking strength, pounds per inch of width		
Warp direction	20	---
Fill direction	15	---
Specific gravity at 25°C	1.5	1.9
Thickness, inch	0.016	0.021

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3.3 **Workmanship.** The reinforcement shall be visibly free from tears, holes, impurities, non-uniform width and thickness, and other defects which could render the reinforcement unsuitable for its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the tests specified in table II.

4.4.1 Lot. A lot shall consist of all reinforcement rolls subject to inspection at the same time from one of the following:

a. A single batch (see 6.3.1).

b. A continuous process (see 6.3.2).

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c. A uniform blend of batches or a uniform blend of portions from a continuous process.

TABLE II. Quality conformance inspections.

Tests and Examinations	Requirement	Test Method
Visual examination	3.3 and section 5	4.5.1
Carbon assay and moisture content	3.2	4.5.2
Ash content	3.2	4.5.3
Weight	3.2	4.5.4
Breaking strength	3.2	4.5.5
Specific gravity	3.2	4.5.6
Thickness	3.2	4.5.7

4.4.2 Sampling. Except where 100 percent sampling is indicated, the number of rolls to be selected shall be in accordance with table III.

TABLE III. Sampling.

Number of Rolls in Lot	Number of Rolls to be Selected
1 - 3	All
4 - 40	3
41 - 65	4

4.4.3 Selection of reinforcement for test. The reinforcement shall be selected from the exposed end of each sampled roll in sufficient quantity to perform all tests specified herein.

4.4.4 Test specimens. Three specimens shall be tested from each sample for each of the properties requiring test specimens. The average test results of three test specimens shall be recorded in addition to recording the individual test specimen value. The average results of each test shall meet the requirements of table I.

4.5 Inspection methods. The following inspection methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes,

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temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 Visual examination. Visual examination shall be made to verify conformance to the requirements of 3.3 and section 5.

4.5.2 Carbon assay and moisture content. The carbon assay determination and moisture content shall be in accordance with the following:

a. Specimen shall be dried to constant weight at 225 ± 5°F before combustion, and percent carbon to the nearest 0.1 percent shall be based on weight of dried specimen.

b. Carbon assay shall be conducted in accordance with ASTM D 3176.

c. Moisture content shall be calculated in accordance with the following:

$$\text{Percent moisture} = \frac{(W - W1)}{W} \times 100$$

Where:

W = specimen weight prior to drying, g

W1 = specimen weight after drying, g

d. Record results to the nearest 0.1 percent.

4.5.3 Ash content. The ash content shall be determined as follows:

a. Dry 3.0 ± 0.6 g of carbon reinforcement in an air-circulating oven at 225 ± 5°F for a minimum of 1 hour.

b. Remove specimen, place in desiccator and cool to ambient temperature.

c. Weigh 2.0 ± 0.4 g (W) of cooled specimen into a tared crucible (B) and place into a muffle furnace at 1400 ± 25°F for a minimum of 3 hours. Weights (W) and (B) shall be measured to the nearest 0.001 g.

d. Raise furnace temperature at end of 3 hours to 1600 ± 25°F for a minimum of 1 hour.

e. Allow an additional 30 minutes burning time or until a constant weight is obtained, should ashing be incomplete.

f. Remove specimen and crucible, and place in a desiccator to cool.

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g. Obtain post-heat weight of specimen and crucible (A) to nearest 0.001 g, and calculate percent ash as follows:

$$\text{Percent ash} = \frac{(A - B)}{W} \times 100$$

Where:

A = weight of crucible and ash, g
B = weight of crucible, g
W = weight of specimen, g

h. Record ash content to the nearest 0.1 percent.

4.5.4 **Weight.** The weight of the material shall be determined in accordance with ASTM D 3776.

4.5.5 **Breaking strength.** The breaking strength test shall be conducted as specified in ASTM D 1682.

4.5.6 **Specific gravity.** Specific gravity shall be determined in accordance with ASTM C 135.

4.5.7 **Thickness.** Thickness shall be determined in accordance with ASTM D 1777 at 5 ± 4 pounds per square inch (psi).

4.6 **Inspection of packaging.** The reinforcement containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The reinforcement shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

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a. Title, number, and date of this specification

b. Manufacturer's name and address

c. Date of manufacture

d. Net weight of contents and number of rolls

e. Acceptance lot number and date of quality conformance inspection

f. Mark each roll with supplier name, lot number, and material name on inside of spool.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The carbon reinforcement is intended for use in the manufacture of carbon cloth phenolic for reinforced plastic nozzle components for the Standard Missile Mk 104 Rocket Motor.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

c. Whether first article inspection is required (see 3.1 and 6.4)

d. Special marking, if other than as specified in 5.2.2

6.3 **Definitions.**

6.3.1 **Batch.** A batch is that amount of reinforcement made as one unit in the manufacturing process.

6.3.2 **Continuous process.** A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the reinforcement is accumulated for the lot.

6.3.3 **Roll.** A single roll of continuous or joined cloth delivered to the impregnator.

6.3.4 **Sample.** A roll of cloth selected in accordance with the requirements of this specification for performance of inspections.

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6.3.5 **Specimen.** A portion of material taken from a sample roll for performance of inspections.

6.4 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Subject term (key word) listing.

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 9330-N024)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-82817	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
REINFORCEMENT, CARBON

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	c. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	d. DATE SUBMITTED (YYMMDD)
e. ADDRESS (Include Zip Code)			

7. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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